Purav Bhatt

Full-Stack & AI Engineer | TypeScript, Node.js, Next.js, Python, Flutter, AWS, LLMs/RAG
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Full-stack engineer focused on reliable backends and production LLM features, from Flutter and Next.js apps to queues and RAG on AWS.

Education

Guru Gobind Singh Indraprastha University — Delhi, India
B.Tech in Artificial Intelligence and Data Science; CGPA: 8.0/10 — *Expected 2027*

Experience

Full-Stack Developer — Sep 2025 – Sep 2026
City Fire Services India — *Delhi, India*

- Built and shipped Marka, the company's attendance and payroll platform, used daily by all 20 employees: two Flutter apps and a Next.js/TypeScript admin portal on Supabase, shipped via PR review and CI.
- Designed the data layer: PostgreSQL with Row-Level Security on every table, REST APIs, Supabase Edge Functions (Deno) and pg_cron jobs, with secrets in Supabase Vault.
- Built GPS-verified check-in with biometrics, a live crew map (Mapbox), on-device OCR for expenses (Google ML Kit), leave approvals and auto-calculated salary slips; fixed GPS race conditions with cancellable async fetches.

Data Science Intern — Jul 2025 – Oct 2025
IIIT Delhi — *Delhi, India*

- Built preprocessing pipelines that turned raw tabular and graph-structured datasets into model-ready inputs, and set up scikit-learn baselines the research team used as reference points for its graph neural network (GNN) experiments.

Projects

PulseBoard | *Next.js, Node.js, Socket.io, PostgreSQL, pgvector, Redis, AWS Bedrock, CDK* | live

- Real-time collaborative Kanban (100 concurrent WebSocket users at 78.5 ms p95 on two 0.25-vCPU tasks), deployed to AWS ECS Fargate, RDS and CloudFront with CDK and GitHub Actions over OIDC.
- RAG "Ask the board" with cited answers and hybrid search (pgvector + full-text, Reciprocal Rank Fusion): Recall@5 0.957, citation precision 1.0, 10/10 prompt-injection cases resisted (instruction override, secret exfiltration).
- LLM evals that gate CI on regressions, AI triage validated with zod, and an XGBoost completion-time model (MAE 16.7 h vs. 26.7 h baseline).

RelayQ | *TypeScript, Node.js, BullMQ, Redis, PostgreSQL, Docker, OpenTelemetry, AWS* | live

- Multi-tenant job service: 33,424 jobs/min on one laptop (3 workers × 5 concurrency, k6); 0 lost or duplicated across 10k jobs with 10 worker kills, via a transactional outbox and idempotent effect log.
- LLM jobs with per-tenant budgets, fallback models and a cost ledger; retries, dead-letter queue, rate limits, HMAC webhooks; traced with OpenTelemetry.

Urban Policy Simulation | *Python, Mesa, OSMnx, FastAPI, Next.js*

- Team project: built the agent-based simulation core (5,000 commuter agents, 8 travel modes) on Delhi's real road network, with a BPR congestion model calibrated to RITES mode-share and DMRC ridership data.

Technical Skills

Languages: TypeScript, JavaScript, Python, Dart, SQL

Frameworks: React, Next.js, Node.js, Express, Flutter, Riverpod, FastAPI, Tailwind CSS, REST APIs, WebSockets (Socket.io)

Data: PostgreSQL, Prisma, Redis, BullMQ, Supabase, pgvector

AI & ML: LLMs, RAG, embeddings, semantic search, LLM evals, Amazon Bedrock, Anthropic API, XGBoost, scikit-learn, pandas, GNNs

Cloud & DevOps: AWS (ECS Fargate, RDS, ElastiCache, S3, CloudFront, CDK), Docker, Git, CI/CD (GitHub Actions), OpenTelemetry, Prometheus, Vitest, Playwright